L1b: Time Value of Money, Abstract Assets, and Net Present Value

CHEME 5660 Cornell University

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Objectives for Today

Today we compare cash flows at different dates, derive accumulation and discount factors, and use them to value an abstract asset.

Today's concepts

- The time value of money: cash flows at different dates cannot be added until they are expressed at the same valuation date.
- An abstract asset: a sequence of dated signed cash flows. The same representation covers a bond, a stock, a factory, or a car.
- Net present value: discount each signed cash flow to the valuation date, then add. Its sign is a statement about the rate you declared, not a guarantee of profit.

Companion notes: L1b lecture notes and notebooks

Lectures and Examples

Lecture: CHEME-5660-L1b-Lecture-TimeValueMoney-Fall-2026.ipynb

Example: CHEME-5660-L1b-NetPresentValue-Fall-2026-WorkedExample.ipynb

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Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Time Value of Money

A dollar today and a dollar at a future date are not directly comparable. Before we compare or add them, we must express both at the same valuation date using a stated conversion rule.

Let P be an amount held at time 0, and let y be a quoted nominal annual yield used for valuation. With one compounding interval per year, the amount after one year is given by:

$$F_1 = \underbrace{(1+y)}_{\mathcal{D}_{1,0}(y;1)} P.$$

For $y > 0$ the factor $\mathcal{D}_{1,0}(y;1) > 1$ moves time-0 dollars forward and is the forward accumulation factor; its inverse $\mathcal{D}_{1,0}^{-1}(y;1) < 1$ moves future dollars back and is the discount factor.

Careful: a later dollar is not *automatically* worth less. That depends on the rate y you declare.

Discrete Compounding and Accumulation Factors

Let P be an amount held at time 0, let $j = 0, 1, 2, \ldots$ index the compounding periods, and let $i_\ell > -1$ be the effective rate earned over the interval $\ell \to \ell + 1$. Then the amount F_j after j periods is given by:

$$F_j = \underbrace{\left[\prod_{\ell=0}^{j-1}(1 + i_\ell)\right]}_{\mathcal{D}_{j,0}(\mathbf{i})} P.$$

If the same nominal annual yield y applies in every interval, with n intervals per year, then $i_\ell = y/n$ for every ℓ and the product collapses to:

$$\underbrace{\mathcal{D}_{j,0}(y;n) = \left(1 + \frac{y}{n}\right)^{j}}_{\text{accumulation}}, \qquad \underbrace{\mathcal{D}^{-1}_{j,0}(y;n) = \left(1 + \frac{y}{n}\right)^{-j}}_{\text{discount}}.$$

Careful: a flat y is an *assumption*. In L2b, y varies with the date.

Continuous Compounding

Let $g(t)$ be a continuously compounded growth rate, carrying units of inverse time, and let T be the horizon. The forward accumulation factor over $0 \to T$ is given by:

$$\mathcal{D}_{T,0}(g) = \exp\left(\int_0^T g(u)\,du\right),$$

which for constant g collapses to:

$$\underbrace{\mathcal{D}_{T,0}(g) = e^{gT},}_{\text{accumulation}} \qquad \underbrace{\mathcal{D}_{T,0}^{-1}(g) = e^{-gT}}_{\text{discount}}.$$

A nominal annual yield y compounded n times per year has the continuously compounded equivalent $g_y := n\log\left(1 + y/n\right)$, so $\left(1 + y/n\right)^{nT} = e^{g_y T}$: these are two representations of one accumulation rule, not two different rules.

Log return: the dimensionless product $r = (\Delta t)\, g$. L2a and L3a build on it.

Who Sets the Valuation Rate?

A valuation rate is not universal. It depends on the cash flow, market convention, horizon, and the risks being modeled: no single policy rate prices every asset.

The Federal Reserve

- Steers short-term rates through interest on reserve balances (IORB), supported by the overnight reverse repo facility; open market operations manage the supply of reserves.
- The FOMC sets a target range for the federal funds rate, the overnight rate on unsecured reserve balances traded between eligible institutions.
- Other market rates respond to policy and market conditions, but are not mechanically pegged to the federal funds rate.

Link: FOMC calendar and meeting materials

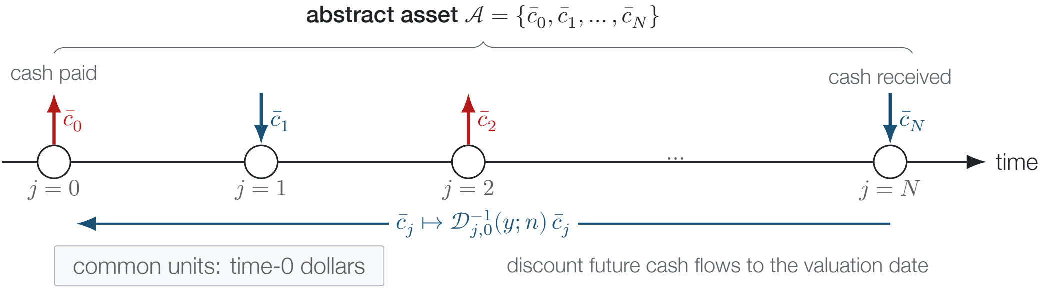
What Valuation Rate Should I Use?

There is no universal answer, and this lecture does not need one: here the yield y is a declared model input. Choosing it well means matching the rate to the cash flow: same currency, same horizon, and an explicit statement of any adjustment made for credit, liquidity, or risk.

Two cautions we rely on later

- A quoted bond yield and an estimated equity required return are different kinds of number. A bond yield is the discount rate that matches price to promised cash flows; an equity required return is an estimate of future compensation for risk.
- Investors expect compensation for systematic risk: the risk that diversification cannot remove. Taking more diversifiable risk does not by itself raise expected return, and an expected return is not a realized return.

An Abstract Asset



Net Cash Flows

An abstract asset is a sequence of dated cash flows denominated in a currency. At cash-flow index j, let $\mathbf{c}_j \in \mathbb{R}^m_{\geq 0}$ collect m nonnegative magnitudes and let $\boldsymbol{\nu}_j \in \{-1, +1\}^m$ give their directions, where $+1$ denotes an inflow and -1 an outflow. The signed net cash flow at index j is given by:

$$\bar{c}_j = \langle \mathbf{c}_j, \boldsymbol{\nu}_j \rangle = \sum_{i=1}^{m} c_{j,i}\, \nu_{j,i}.$$

Over the dates $j = 0, 1, \dots, N$, the undiscounted total $\sum_{j=0}^{N} \bar{c}_j$ is not a time-0 value: its terms are dollars from different dates.

Next: discount every term to a common date, and the result is net present value.

Net Present Value

Let $\bar{c}_j$ be the signed net cash flow on date j, positive for an inflow and negative for an outflow, over the dates $j = 0, 1, \dots, N$. Discount each one to the valuation date with $\mathcal{D}^{-1}_{j,0}(y;n)$, then add. The net present value at time 0 is given by:

$$\text{NPV}_0(y) = \sum_{j=0}^{N} \mathcal{D}^{-1}_{j,0}(y;n)\,\bar{c}_j = \underbrace{\left\langle \mathcal{D}^{-1}_{\star,0}(y;n),\ \bar{\mathbf{c}} \right\rangle}_{\text{time-0 dollars}}.$$

The $\star$ runs over every cash-flow index, so $\mathcal{D}^{-1}_{\star,0}(y;n)$ and $\bar{\mathbf{c}}$ are ordered $(N+1)$-vectors and NPV is a single scalar product, which is exactly how we compute it in code.

Worked example: ten years of Model S ownership, $\text{NPV}_0 = -\$121{,}484$

Summary

This lecture introduced time-value-of-money calculations and the representation of investments as dated cash flows.

Key takeaways

- **Dated dollars must be compared at a common valuation date.** A forward accumulation factor carries value forward and its inverse discounts it back, so cash flows can be added only once they share a date.
- **Rate conventions must be stated.** A nominal yield quoted with a compounding frequency and its continuously compounded equivalent describe one accumulation rule, so the convention travels with the number.
- **NPV values an asset relative to a declared rate.** The sign of the result is a statement about the cash-flow model and the discount rate you declared.

These ideas provide the valuation framework used throughout the course.